My research interests lie in visualization, accessibility, and natural language processing (NLP), particularly at the intersection of model interpretability and data visualization accessibility. While alt text online is rare and often low-quality, vision language model (VLM)-generated image descriptions for blind and low vision (BLV) users can introduce deceptive factual errors and reproduce harmful stereotypes. I believe understanding how these generations occur and the broader impact they have on disabled people are both important parts of improving the accessibility of visualizations at scale. Building transparent models is particularly important in accessibility technology (AT) to give disabled users agency over black-box outputs, but many interpretability methods remain inaccessible. I plan to pursue these problems in my PhD by:

(**a**) developing visualization systems to interpret LLMs,

(**b**) helping improve BLV accessibility of data visualizations on the web, and

(**c**) creating tools to increase the transparency of LLMs in accessibility technology.

Visualization for model interpretability

I am interested in developing visualization systems to interpret language models at scale. In my first research project at the University of Washington, I helped create interactive visualization tools to analyze recorded neuron activity in mice. These systems were used by an international group of neuroscience labs to promote their datasets [1, 2] (*two bioRxiv preprints, one in submission to Nature*) and were later adapted into software to help researchers plan probe insertions [3]. Making sense of seemingly random neuron firing made me curious about visually interpreting other complex open-weight systems.

This informed my work in Noah Smith's NLP group, where I led a project showing that language models finetuned on data from different times are arranged intuitively in weight space [4] (*first author paper at ACL 2024*). For example, a model trained on 2012 data will have weights that are more similar to a 2013 model than a 2016 model. Using this structure, I implemented methods to update models to new time periods by interpolating between their weights. In our analysis, I was drawn to the challenge of displaying relationships between high-dimensional model components—connecting visualization and mechanistic interpretability. Most research on LLMs through this lens focuses on embedding or attention layers [5], but I believe we should develop systems to view the *whole* model during and after training. In our paper, for instance, we found that feedforward layers are also arranged in a visually intuitive way based on time and often play the largest role in year-to-year model degradation. While I found this analysis interesting and believe it can shed light on how models work globally, the post-hoc nature of our work made me realize that I want to focus on more *practical* interpretability projects. Moving away from static explanations, how can we visualize LLM weights during everyday use to interpret their generations?

Improving the accessibility of data visualizations

Another one of my research goals is to improve the accessibility of visualization tools. For BLV developers, many data visualization systems and academic publications are still imperceptible to screen readers [6-8]. To help address this issue in computational notebooks, I developed a Python package called MatplotAlt [9] that provides functions to automatically generate VLM or template-based Matplotlib figure alt text, and then embed or export captions in the notebook environment (*first author paper in submission to EuroVis 2025*). Working on this project introduced me to the scope of barriers in ubiquitous software—Jupyter, for instance, has no interface for embedding text in figures—and motivates me to continue improving the accessibility of visualization and NLP systems. Like interpretability, however, I believe

retrofitting access to existing visualizations is not enough. I plan to search for novel ways to understand and interact with data without relying on visual representations.

Evaluating alt texts generated by MatplotAlt was also challenging. Even if there are ground truth figure descriptions written by humans, similarity-based comparisons miss that readers may prefer different features and levels of detail. At the same time, evaluating alt texts on the web at scale is infeasible with only manual annotation. In my PhD, I'm interested in creating new tools to evaluate figure descriptions based on customizable success criteria.

Accessible interpretability and transparent models in AT

Bridging my research interests, I believe there are unexplored opportunities at the intersection of accessibility and interpretability. Model transparency is particularly important to give disabled people agency over AT that uses black-box LLMs. Even if VLMs can generate descriptive alt text, there is no way to fully evaluate its correctness without perceiving the image, and the extent to which BLV users trust these captions is still unknown. Incorporating tools to explain why models describe chart features and context could help BLV users determine whether to trust LLM-generated alt texts. I believe accessible interpretability systems are also more rigorous and robust. Saliency maps, for example, rely on the assumption that model behavior is aligned to visual areas of focus, and are notoriously misleading without extra context. More accessible textual descriptions could explicitly list feature scores and uncertainty for image regions based on natural language instructions.

Perceptions of LLMs have a large impact on how they are used for accessibility, and whether interpretability tools are used in practice. In my senior thesis for comparative history, I wrote about how arguments humanizing ChatGPT in the 2023 Senate Oversight of AI hearing helped obscure exploitative crowdworking conditions. For BLV users, the complexities of relying on black-box models are similarly overshadowed by claims that they can summarize content in "the way the sighted naturally scan them" [10]. In future work with the humanities, I'm interested in decoupling these narratives from applications of NLP and visualization in AT.

Personal Statement and Future Goals

After starting college early, I wanted to explore as many different subjects as possible, which led me to pursue visualization, NLP, accessibility, and the humanities. I'm motivated to continue my graduate studies by a desire to deepen my understanding of these fields, and my positive experiences conducting research. I believe academia is the best place to pursue the interdisciplinary work I find most fulfilling, with access to a wide range of departments and faculty mentors. I've also spent my time in Seattle volunteering to help connect homeless youth to resources, which has made me value meeting people where they are rather than focusing on speculative technological fixes. My goal is to continue improving access to resources in my future work, both through human-centered accessibility projects and teaching. My career plan is to become a professor and lead a research group. A PhD in CS / Informatics at [X university] will allow me to pursue this goal and grow into a more experienced researcher and educator.

[1] **International Brain Laboratory** *et al*., "A brain-wide map of neural activity during complex behaviour," *bioRxiv,* 2023.

[2] C. Findling, **International Brain Laboratory**, *et al.*, "Brain-wide representations of prior information in mouse decision-making," *bioRxiv*, 2023.

[3] D. Birman *et al.*, "Pinpoint: trajectory planning for multi-probe electrophysiology and injections in an interactive web-based 3D environment," *bioRxiv*, 2023.

[4] **K. Nylund**, S. Gururangan, and N. Smith, "Time is encoded in the weights of finetuned language models," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 2571–2587.

[5] H. Zhao *et al.*, "Explainability for large language models: A survey," *ACM Trans. Intell. Syst. Technol.*, vol. 15, no. 2, pp. 1–38, 2024.

[6] A. Srinivasan *et al*., "Azimuth: Designing Accessible Dashboards for Screen Reader Users." *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*, 2023.

[7] L. L. Wang *et al.*, "Improving the accessibility of scientific documents: Current state, user needs, and a system solution to enhance scientific PDF accessibility for blind and low vision users," *arXiv [cs.DL]*, 2021.

[8] V. Potluri *et al*., "Notably Inaccessible—Data Driven Understanding of Data Science Notebook (In) Accessibility." *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*, 2023.

[9] **K. Nylund**, J. Mankoff, and V. Potluri. "MatplotAlt: A Python Library for Adding Alt Text to Matplotlib Figures in Computational Notebooks," *Under review at EuroVis 2025*.

[10] OpenAI, "Transforming visual accessibility," https://openai.com/index/be-my-eyes, Accessed November 27, 2024.